

Newton's method



Outline

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Newton's method

Introduction: Newton's method

- ▶ In this section we see how to numerically approximate the solutions of an equation $f(x) = 0$
- ▶ The idea is to repeatedly replace f with its tangent line, building a recursively defined sequence
- ▶ The theorem on Newton's method gives the hypotheses that guarantee convergence

- Suppose we want to solve the equation

$$f(x) = 0$$

which is equivalent to looking for the intersections of the graph of f with the x -axis, or the zeros of the function.

Suppose moreover: i) that the solution $x = c$ exists, is unique and lies inside the interval $[a, b]$; ii) that the function is differentiable in $[a, b]$; iii) that $f'(x) \leq 0$ (decreasing function) and $f''(x) \geq 0$ (convex function) for every $x \in [a, b]$.

- Let us then start from $x_0 = a$ and linearize the equation $f(x) = 0$ by replacing f with the tangent line to its graph at the point $(x_0, f(x_0))$. This line has equation:

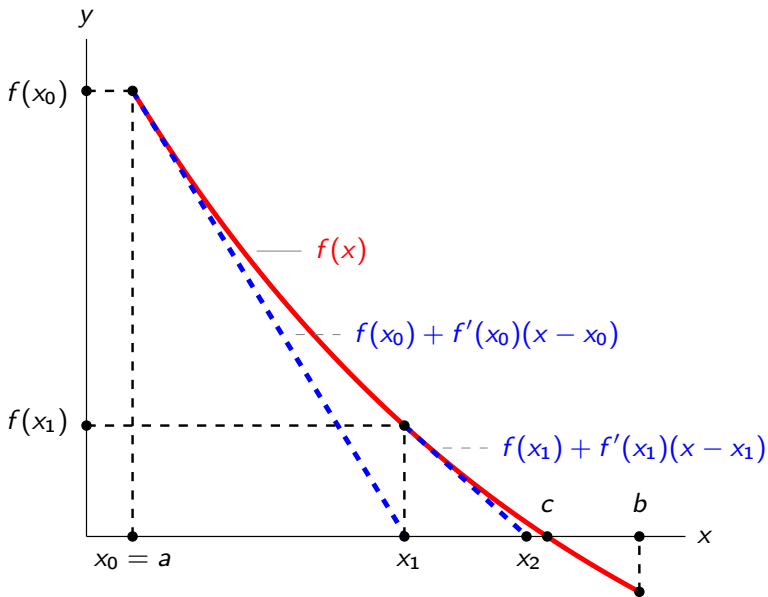
$$y = f(x_0) + f'(x_0)(x - x_0)$$

Instead of solving $f(x) = 0$, we solve:

$$f(x_0) + f'(x_0)(x - x_0) = 0$$

calling the solution x_1 we obtain:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}, \quad \text{the first approximation of } c$$



- Let us now start from x_1 and linearize as before with the tangent line at the new point $(x_1, f(x_1))$. This line has equation:

$$y = f(x_1) + f'(x_1)(x - x_1)$$

As before, we solve:

$$f(x_1) + f'(x_1)(x - x_1) = 0$$

obtaining

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}, \quad \text{the second approximation of } c$$

Proceeding with x_2 as with x_1 and continuing, we obtain the following recursively defined sequence:

$$x_0 = a, \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \text{with } n \in \mathbb{N}$$

In this way, the term x_n can be constructed starting from x_0 with n iterations of the same algorithm. The method is therefore very well suited to automatic computation.

- ▶ The main idea of this method, called **Newton's method**, is therefore to construct a sequence that, **under certain hypotheses**, converges to c .

Theorem 1: Newton's method

Let $f : [a, b] \rightarrow \mathbb{R}$ be twice differentiable in $[a, b]$. If the following three hypotheses hold:

1. $f(a) \cdot f(b) < 0$
2. $f'(x)$ and $f''(x)$ have constant sign in $[a, b]$
3. $f(a) \cdot f''(a) > 0$

then there exists one and only one point $c \in (a, b)$ such that $f(c) = 0$, and the sequence

$$x_0 = a, \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

converges to c from below.

Theorem 1: Newton's method (continued)

If, instead of hypothesis 3, the following hypothesis holds

$$3'. f(b) \cdot f''(b) > 0$$

then the sequence

$$x_0 = b, \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

converges to c from above.

- ▶ Note that, under hypotheses 1. and 2., either 3. or 3'. is certainly satisfied. The function f'' has the same sign at a and b by 2., and the function f has opposite signs at a and b by 1. Therefore either 3. or 3'. holds.
- ▶ This means that the method is applicable whenever hypotheses 1. and 2. are satisfied: we only need to choose appropriately whether to set $x_0 = a$ or $x_0 = b$.
- ▶ Condition 2, apparently restrictive, will in general be satisfied provided we choose an interval $[a, b]$ small enough.

Proof (1/6)

Since f is continuous, being differentiable in $[a, b]$, and 1 holds, by the intermediate zero theorem there exists at least one $c \in (a, b)$ such that $f(c) = 0$.

Moreover, since $f'(x)$ has constant sign in (a, b) , f is strictly monotone, hence this point c is unique (a strictly monotone function cannot vanish at two distinct points).

This proves the existence and uniqueness of the point c at which f vanishes. We will now prove that the sequence is monotone. From this it will follow that the sequence is convergent, by the theorem on monotone sequences.

Proof (2/6)

We start by showing that if the sequence x_n converges, then $x_n \rightarrow c$. If $x_n \rightarrow \ell \in [a, b]$, passing to the limit in the equality

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \text{we get} \quad \ell = \ell - \frac{f(\ell)}{f'(\ell)}$$

(we used the fact that f is continuous, and f' is also continuous, since f' is differentiable, because f'' exists by hypothesis). From the last equality it follows that $f(\ell) = 0$, so that $\ell = c$ (the only point where the function vanishes).

Proof (3/6)

Let us therefore prove that x_n is monotone, under hypotheses 1, 2, 3. Without loss of generality, we assume $f(a) < 0$ and, by 1, consequently $f(b) > 0$. Since, by 2, $f'(x)$ has constant sign in $[a, b]$, we must have $f'(x) > 0$ on all of $[a, b]$ (if the other inequality held, f would be decreasing, and we could not have $f(a) < 0 < f(b)$). Since $f(a) < 0$ and 3 holds, $f''(a) < 0$; hence, since by 2 $f''(x)$ has constant sign, $f''(x) < 0$ on all of $[a, b]$. **We are therefore proving the case of an increasing and concave function.** The other cases are proved analogously.

Proof (4/6)

Consider the function:

$$g(x) = x - \frac{f(x)}{f'(x)},$$

we have $x_{n+1} = g(x_n)$ and $g(c) = c$ since $f(c) = 0$

We have:

$$g'(x) = 1 - \frac{f'(x)^2 - f(x)f''(x)}{f'(x)^2} = \frac{f(x)f''(x)}{f'(x)^2} > 0 \iff f(x) < 0$$

because $f''(x) < 0$ on all of $[a, b]$. Hence $g'(x) > 0$ for $x \in [a, c]$, i.e., on $[a, c]$ the function g is strictly increasing.

Proof (5/6)

Keeping these facts in mind, we now prove that:

$$x_n < x_{n+1} < c, \quad \forall n \in \mathbb{N} \quad (1)$$

For $n = 0$ we have

$$x_1 = a - \frac{f(a)}{f'(a)} > a$$

because $f(a) < 0$ and $f'(a) > 0$, hence $x_1 > x_0$. Moreover $a < c$ (and $x_0 = a$), and since g is increasing, this implies:

$$g(a) < g(c) \quad \text{that is} \quad x_1 < c$$

Hence we have proved that

$$x_0 < x_1 < c$$

Proof (6/6)

Applying now g to the previous inequalities (g is strictly increasing on $[a, c]$, and the three points lie in this interval, hence g preserves the inequalities), we have

$$g(x_0) < g(x_1) < g(c) \quad \text{that is} \quad x_1 < x_2 < c$$

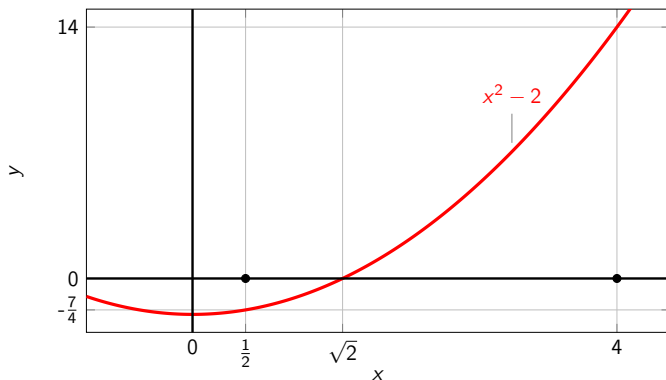
Applying g again, we find $x_2 < x_3 < c$, and so on. Therefore (1) is true, and in particular x_n is monotone. This concludes the proof.



Example (1/3)

Consider the function:

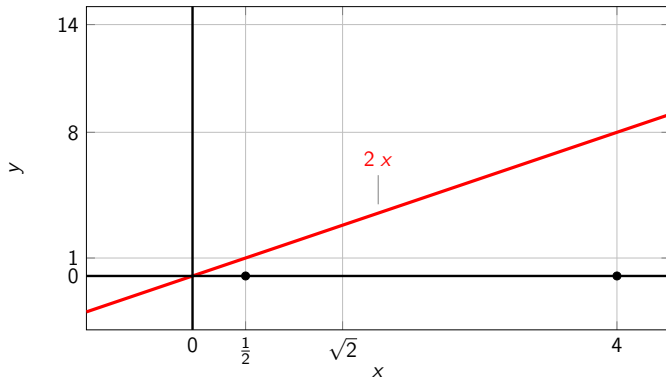
$$f(x) = x^2 - 2, \quad f'(x) = 2x, \quad f''(x) = 2$$



Example (2/3)

We have:

$$f\left(\frac{1}{2}\right) = -\frac{7}{4} < 0 \quad \text{and} \quad f(4) = 14 > 0$$



Example (3/3)

Consider the interval $\left[\frac{1}{2}, 4\right]$; we have:

$$f'(x) > 0 \quad \text{and} \quad f''(x) > 0, \quad \forall x \in \left[\frac{1}{2}, 4\right]$$

Example (1/3)

Hence on the interval $[\frac{1}{2}, 4]$ the function $f(x) = x^2 - 2$ satisfies the hypotheses of the theorem.

We are in the case of a function that is strictly increasing on $[\frac{1}{2}, 4]$, since $f'(x) > 0, \forall x \in [\frac{1}{2}, 4]$, and convex on $[\frac{1}{2}, 4]$, since $f''(x) > 0, \forall x \in [\frac{1}{2}, 4]$. The endpoints of the chosen interval are $a = \frac{1}{2}$ and $b = 4$.

Hypothesis 3' holds, that is, $f(4) \cdot f''(4) > 0$, and the sequence becomes:

$$x_0 = 4, \quad x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right) \quad \text{with } n \in \mathbb{N}$$

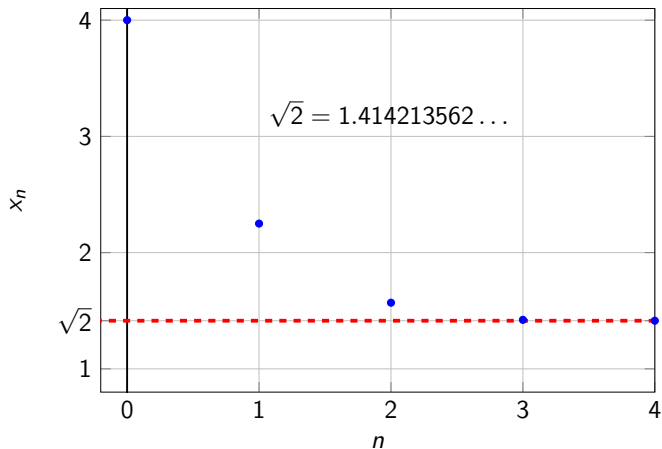
Example (2/3)

Computing the first values of the sequence we have:

$$x_0 = 4, \quad x_1 = \frac{9}{4} = 2.25, \quad x_2 = \frac{113}{72} = 1.569444444 \dots$$

$$x_3 = \frac{23,137}{16,272} = 1.421890363 \dots, \quad x_4 = \frac{1,064,876,737}{752,970,528} = 1.414234285 \dots$$

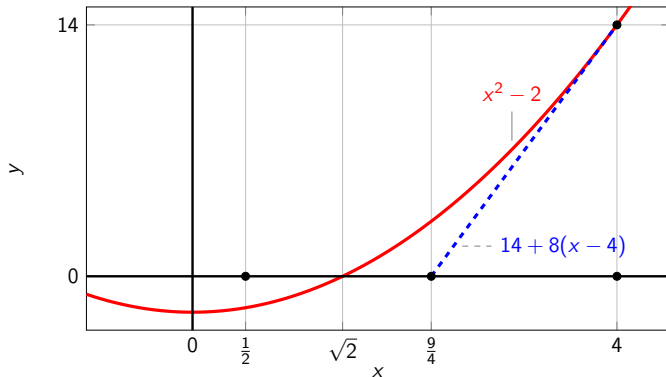
Example (3/3)



Example (1/2)

At the first iteration the tangent line is:

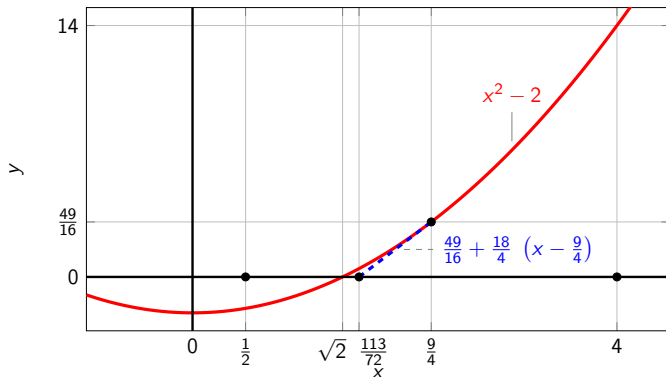
$$y = 14 + 8(x - 4) \quad \text{and} \quad x_1 = \frac{9}{4}$$



Example (2/2)

At the second iteration the tangent line is:

$$y = \frac{49}{16} + \frac{18}{4} \left(x - \frac{9}{4} \right) \quad \text{and} \quad x_2 = \frac{113}{72}$$



k -th root (1/2)

- ▶ Newton's method can be used, for example, to compute the k -th root of a number $c \in \mathbb{R}_+$, by looking for the zeros of the following function:

$$f(x) = x^k - c \quad \text{that is} \quad x = \sqrt[k]{c}$$

Given a value $b \geq \sqrt[k]{c}$, we have the recursively defined sequence:

$$x_0 = b, \quad x_{n+1} = x_n - \frac{x_n^k - c}{k x_n^{k-1}} \quad \text{with } n \in \mathbb{N}$$

k -th root (2/2)

expanding we obtain

$$x_n - \frac{x_n^k - c}{k x_n^{k-1}} = \frac{k x_n^k - x_n^k + c}{k x_n^{k-1}} = \frac{1}{k} \left((k-1)x_n + \frac{c}{x_n^{k-1}} \right)$$

and hence the sequence becomes:

$$x_0 = b, \quad x_{n+1} = \frac{1}{k} \left((k-1)x_n + \frac{c}{x_n^{k-1}} \right) \quad \text{with } n \in \mathbb{N}$$

With $k = 2$ we obtain the sequence of Heron's algorithm:

$$x_0 = b, \quad x_{n+1} = \frac{1}{2} \left(x_n + \frac{c}{x_n} \right) \quad \text{with } n \in \mathbb{N}$$

Summary

- ▶ **Newton's method:** $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, i.e., the zero of the tangent line at $(x_n, f(x_n))$
- ▶ **Hypotheses:** $f(a)f(b) < 0$, f' and f'' of constant sign, start from the endpoint where $f \cdot f'' > 0$
- ▶ The sequence is monotone and converges to the unique zero c
- ▶ Application: k -th root and Heron's algorithm

Error estimates for Newton's method



Introduction: Error estimates

- ▶ In this section we estimate how quickly the error of Newton's method tends to zero
- ▶ Using Taylor's formula with Lagrange remainder we show that convergence is quadratic
- ▶ We obtain an error estimate that does not depend on the unknown zero c

- The **absolute error** of Newton's method is:

$$\varepsilon_n = |x_n - c|$$

where $c \in (a, b)$ is the zero of the function, that is, $f(c) = 0$.

The value of c is unknown, hence we need a way to analyze the values of the absolute errors that is independent of c . We also want to understand how “fast” the error decreases by comparing the errors of two consecutive iterations, that is, we want to determine the “**rate of convergence**” of the algorithm.

- Using the theorem on the Taylor formula with Lagrange remainder we can write:

$$f(x) = f(x_n) + f'(x_n)(x - x_n) + \frac{1}{2} f''(\mu)(x - x_n)^2$$

for some μ between x_n and x . In particular, choosing $x = c$ we have:

$$0 = f(c) = f(x_n) + f'(x_n)(c - x_n) + \frac{1}{2} f''(\mu)(c - x_n)^2 \quad (2)$$

Recalling that x_{n+1} was obtained as the solution of the equation:

$$f(x_n) + f'(x_n)(x - x_n) = 0$$

we have

$$0 = f(x_n) + f'(x_n)(x_{n+1} - x_n) \quad (3)$$

Subtracting (3) from (2), we obtain:

$$0 = f'(x_n)(c - x_{n+1}) + \frac{1}{2} f''(\mu) (c - x_n)^2$$

which in turn implies:

$$x_{n+1} - c = \frac{1}{2} \frac{f''(\mu)}{f'(x_n)} (x_n - c)^2 \quad \text{hence} \quad |x_{n+1} - c| = \frac{1}{2} \frac{|f''(\mu)|}{|f'(x_n)|} |x_n - c|^2 \quad (4)$$

Now, if x_n is close to c (that is, $x_n \approx c$), then μ , which lies between x_n and c , is also close to c ; hence in these cases:

$$f''(\mu) \approx f''(c) \quad \text{and} \quad f'(x_n) \approx f'(c)$$

then, substituting, we obtain:

$$|x_{n+1} - c| \approx \frac{1}{2} \frac{|f''(c)|}{|f'(c)|} |x_n - c|^2 \quad \text{hence} \quad \varepsilon_{n+1} \approx \frac{1}{2} \frac{|f''(c)|}{|f'(c)|} \varepsilon_n^2 \quad (5)$$

The absolute error of Newton's method at each step is proportional to the square of the absolute error at the previous step. **The rate of convergence is quadratic.**

- ▶ However, the error is also proportional to the constant:

$$\frac{|f''(c)|}{2 |f'(c)|}$$

hence, if $|f'(c)|$ is very small, or zero, or $|f''(c)|$ is very large, the convergence may be very slow or may even fail to occur.

- ▶ Let us go back to equation (4), and determine $L, M > 0$ such that:

$$|f'(x_n)| \geq L \quad \text{and} \quad |f''(\mu)| \leq M$$

then we can write:

$$|x_{n+1} - c| \leq \frac{1}{2} \frac{M}{L} |x_n - c|^2 \quad \text{hence} \quad \varepsilon_{n+1} \leq \frac{1}{2} \frac{M}{L} \varepsilon_n^2$$

We have obtained the relation between ε_{n+1} and ε_n , which is independent of both x_n and c (but depends on M and L).

Example (1/2)

Let us go back to the previous example:

$$f(x) = x^2 - 2, \quad f'(x) = 2x \quad \text{and} \quad f''(x) = 2$$

Hence we can choose $M = 2$ and, since $x_n \geq 1, \forall n \in \mathbb{N}$, we can choose $L = 2$ (in this case $\frac{M}{L} = \frac{2}{2} = 1$).

We know that $\sqrt{2} > 1$, hence with $n = 2$ we have

$$\varepsilon_2 \leq \left| \frac{113}{72} - 1 \right| \approx 0.56944, \text{ and we have:}$$

Example (2/2)

iter.	x_n	estimate of $\sqrt{2}$	ε_n
$n = 3$	$\frac{23,137}{16,272}$	1.421890363...	$\leq \frac{1}{2} \varepsilon_2^2 \approx 0.16213...$
$n = 4$	$\frac{1,064,876,737}{752,970,528}$	1.414234285...	$\leq \frac{1}{2} \varepsilon_3^2 \approx 0.01314...$

Summary

- ▶ **Absolute error:** $\varepsilon_n = |x_n - c|$
- ▶ $\varepsilon_{n+1} \approx \frac{1}{2} \frac{|f''(c)|}{|f'(c)|} \varepsilon_n^2$: quadratic convergence, slow if $|f'(c)|$ is small or $|f''(c)|$ is large
- ▶ With $|f'| \geq L$ and $|f''| \leq M$: $\varepsilon_{n+1} \leq \frac{1}{2} \frac{M}{L} \varepsilon_n^2$, independent of c

Final summary

- ▶ Newton's method approximates the zeros of f by repeatedly linearizing the function
- ▶ Under suitable hypotheses on the signs of f , f' , f'' the sequence converges monotonically
- ▶ Convergence is quadratic and the error can be estimated with constants M and L